Biological and climatic constraints drive reproductive synchrony

Supplementary figures

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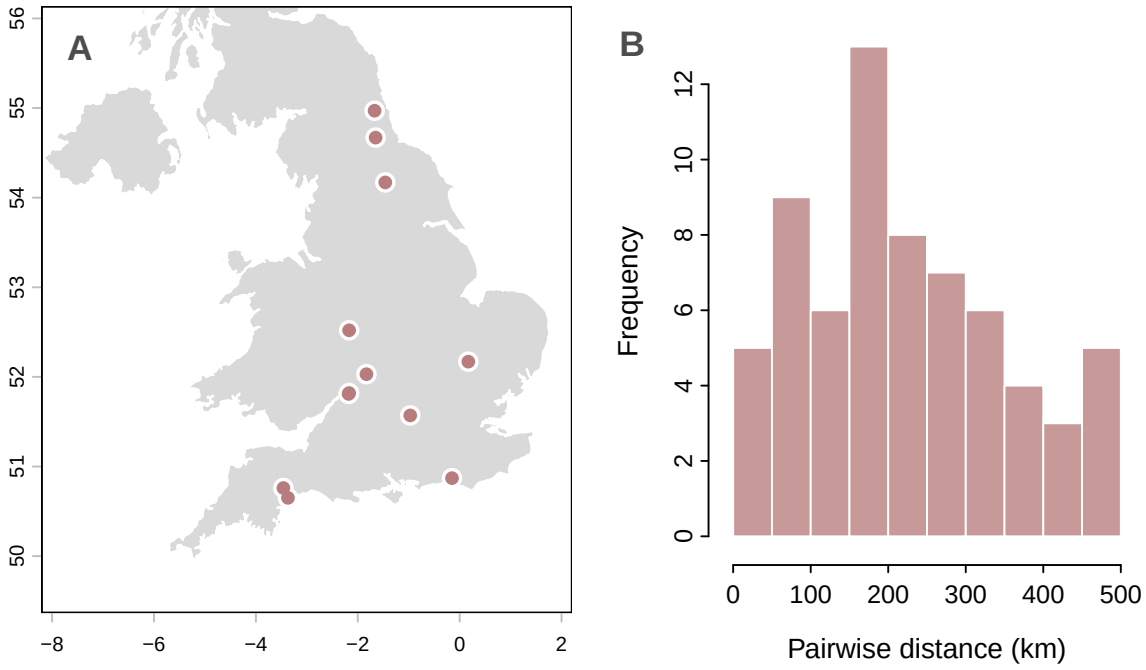


Figure S1: **(A) Map of the beech forest sites (England).** We used data from 171 trees on 12 sites. **(B) Distributions of pairwise distance between sites.** Sites are on average 224 km apart.

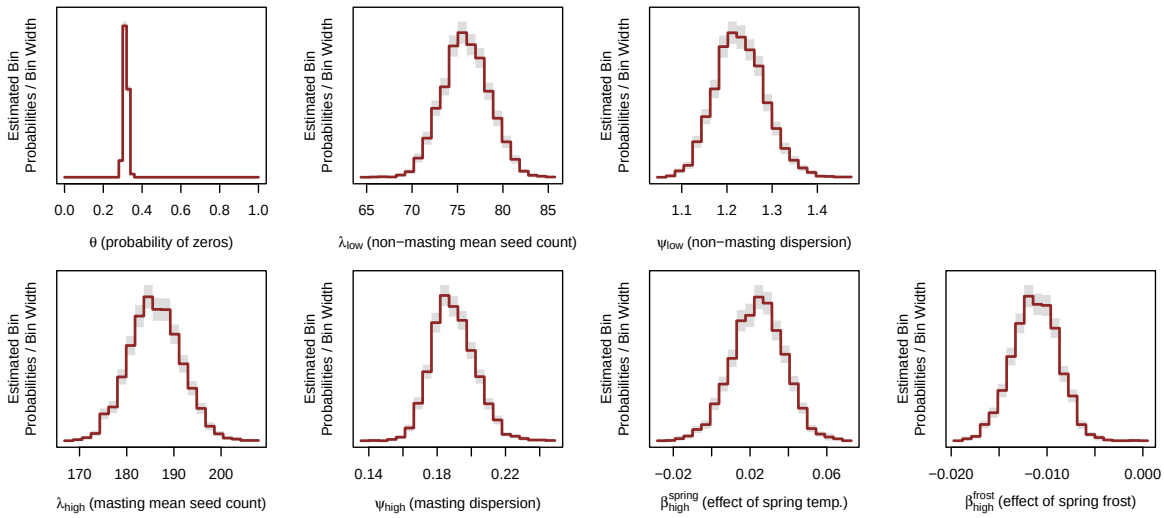


Figure S2: **Posterior distributions of the parameters related to seed count distributions (main text model).** These parameters control the number of seeds produced in low and high states. *Spring temp.* corresponds to the average spring temperature in April and May. *Spring frost* correspond to the growing degree days accumulated until the last frost day. See Methods for details.

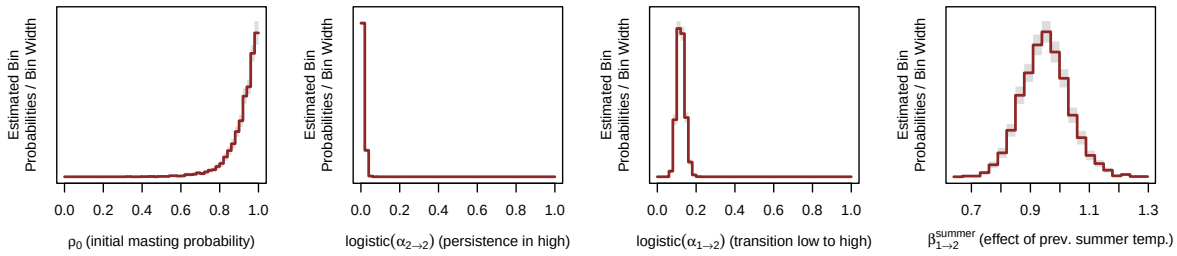


Figure S3: **Posterior distributions of the parameters related to state transitions (main text model).** These parameters control the transition probabilities between states (1 = low, 2 = high). *Prev. summer temp.* corresponds to the average maximum temperature in July and August (of the previous year). See Methods for details.

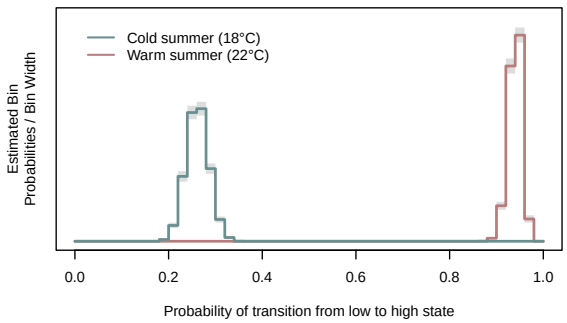


Figure S4: **Posterior distributions of transition probability from low to high reproductive states in a cold vs. a warm summer.** On average, the transition probability during a warm summer is 3.6 times higher than during a cold summer.

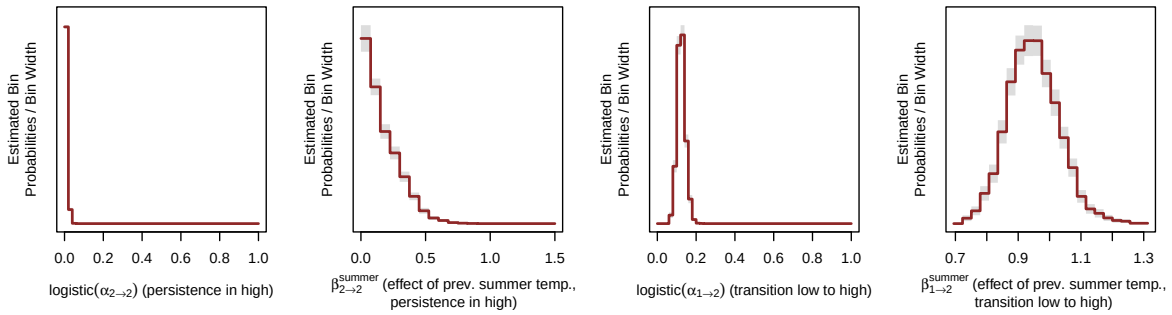


Figure S5: **Posterior distributions of the parameters related to state transitions (model with summer temp. on both transitions).** In this model, we tested whether the persistence in a high state was affected by previous summer temperature ($\beta^{summer}_{2\to2}$). Because this model is more complex (one additional parameter), we constrained both beta parameters to be > 0. This lower bound constraint limits the effect of higher summer temperatures (both previous and current year) to increase seed count as is widely expected (though this constraint was not present in the model presented in the main text and we found the same positive response).

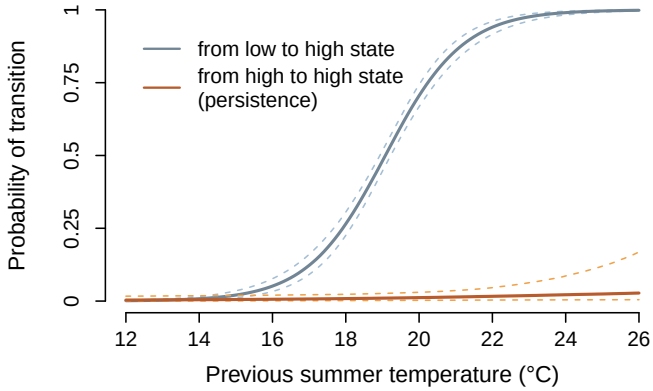


Figure S6: **Effect on previous summer temperature on the probability of transition from a low reproductive to a high reproductive state, and on the probability of persistence in a high reproductive state (model with summer temp. on both transitions, see Supp. Fig S5).** Even when summer temperature could influence both the transition to a high state and the persistence in a high state, trees could not become stuck indefinitely in a high state.

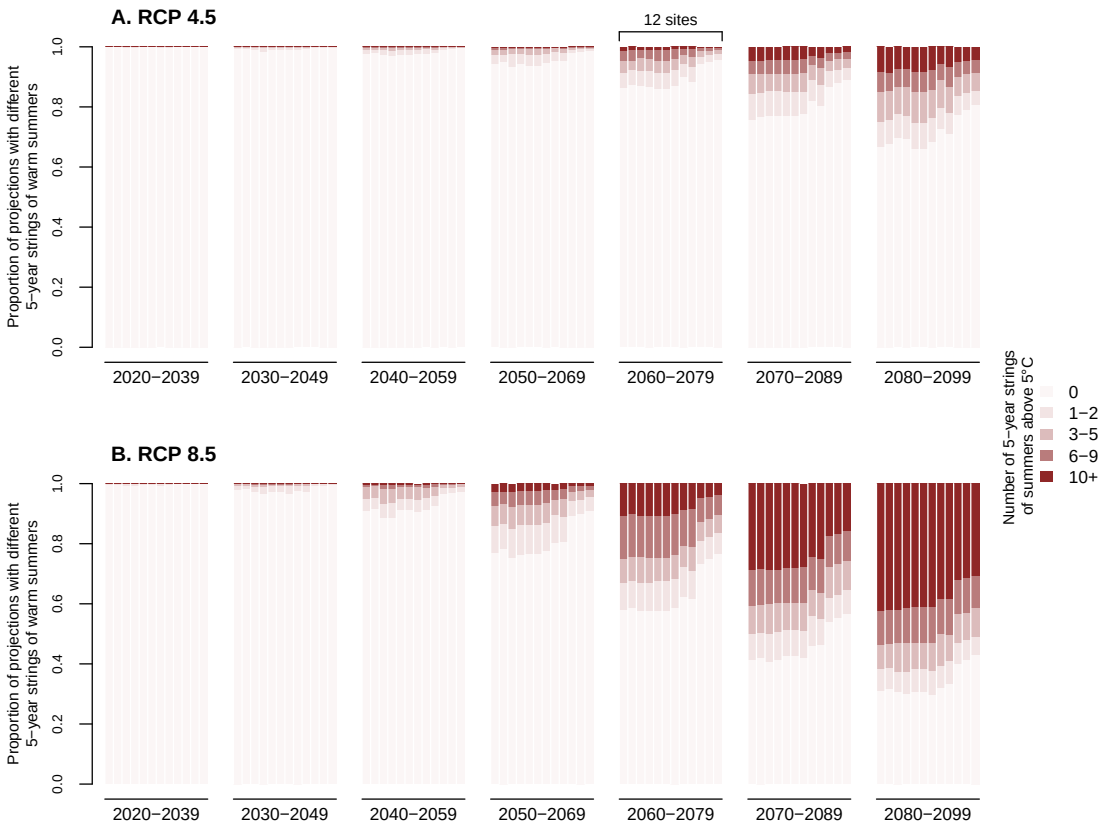


Figure S7: **Projected frequency of strings of warm summers under scenario (A) RCP 4.5 and (B) RCP 8.5.** Each bar shows the proportion of 1000 UKCP18 probabilistic projections (Met Office Hadley Centre, 2018) falling into each category of 5-year strings of consecutive summers at least 5 °C warmer than the 1981-2000 baseline for one site. Strings were counted as 5-year windows and can overlap, so a 20-year period contains a maximum of 16 strings. Within each 20-year period, bars show the 12 sites, ordered by latitude (from south to north). RCP 8.5 represents a high-emission scenario now considered unlikely given current policies and energy trends (Hausfather and Peters, 2020); however, climate models also underestimate recent summer warming and heat extremes in Western Europe (Schumacher et al., 2024).

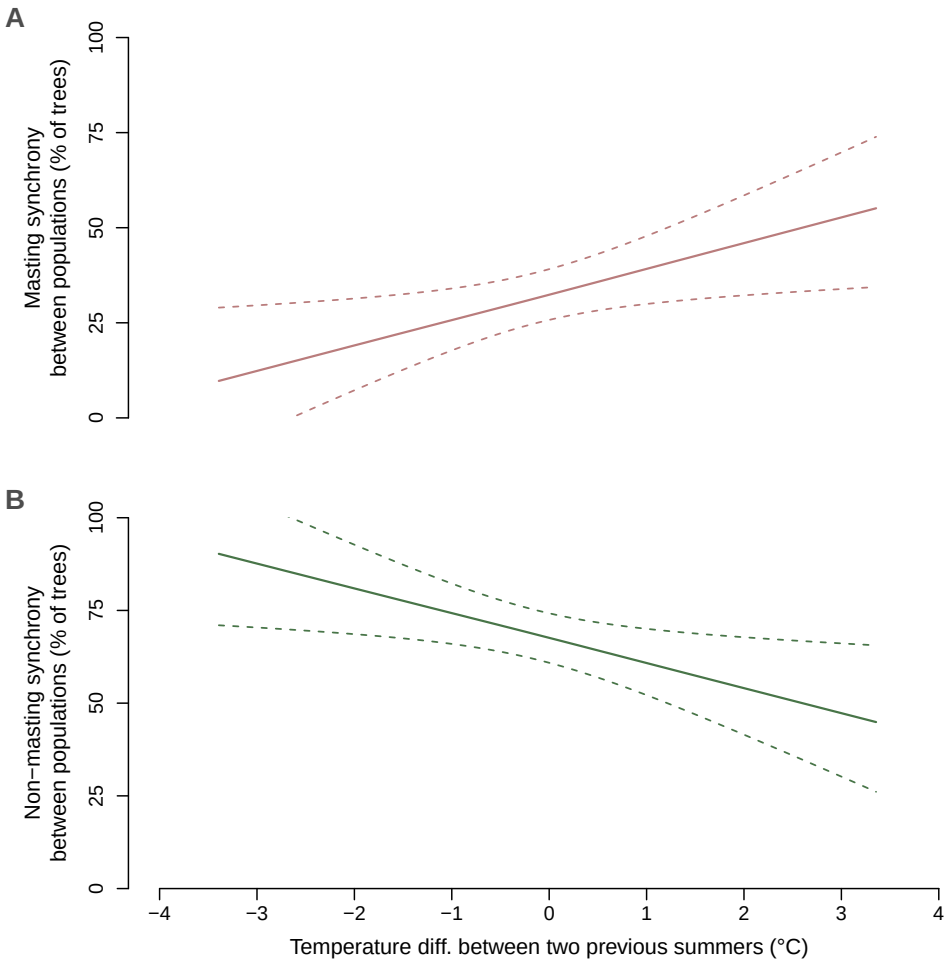


Figure S8: **The interaction of previous-year summer conditions and latent reproductive states (C: high reproductive state, D: low reproductive state) explain the apparent two-year lag effect of climate on observed synchrony across populations.** The solid line represents the median, the dashed lines the 90% uncertainty interval.

References

Met Office Hadley Centre. UKCP18 Probabilistic Climate Projections, Centre for Environmental Data Analysis., 2018. URL http://catalogue.ceda.ac.uk/uuid/9842e395f2d04f48a177c3550756bf98.

Zeke Hausfather and Glen P. Peters. Emissions – the 'business as usual' story is misleading. *Nature*, 577(7792):618–620, January 2020. ISSN 1476-4687. doi: 10.1038/d41586-020-00177-3.

Dominik L. Schumacher, Jitendra Singh, Mathias Hauser, Erich M. Fischer, Martin Wild, and Sonia I. Seneviratne. Exacerbated summer European warming not captured by climate models neglecting long-term aerosol changes. *Communications Earth & Environment*, 5(1), April 2024. ISSN 2662-4435. doi: 10.1038/s43247-024-01332-8.